Causal Obstruction–Response Theory of Transformer Decisions

A Local-to-Global Framework for Decision Formation, Representation Detection, Repair, and Stability

Visya Gregoriy
visyagregoriy@gmail.com

Abstract

Modern language models do not make decisions in a single place. A factual answer, multiple-choice selection, next-token commitment, or tool-use action emerges from distributed computations across tokens, layers, attention heads, MLPs, residual streams, and learned feature directions. Yet many tools for analyzing these decisions focus either on final confidence scores, such as entropy and logit margin, or on locating individual components whose intervention changes the output. These views are useful but incomplete: they do not directly measure whether the model's distributed internal evidence is mutually compatible. We propose *Causal Obstruction–Response Theory*, a local-to-global framework for Transformer decisions. For a candidate contrast $c = (a, b)$, we represent local decision evidence as germs over a computation complex, define compatibility maps between internal sites, and measure failure of local evidence to glue into a stable global decision structure using a corrected obstruction energy. The obstruction is defined metrically as distance to the space of compatible global sections and computed through a weighted sheaf-Laplacian residual, avoiding the invalid construction $[ds] \in H^1$ for a globally defined cochain. We then couple obstruction to a causal response rule that predicts which activation patch, ablation, residual edit, or feature intervention should reduce obstruction and change the candidate margin. The central empirical loop is falsifiable: obstruction predicts intervention sites; predicted interventions change obstruction; obstruction change predicts logit-margin change; and margin change flips, repairs, or stabilizes decisions.

Professor-feedback addition. Before steering, the paper must first define what internal object is being detected or represented. Therefore the framework is organized as

$$\text{representation/detection} \longrightarrow \text{monitoring} \longrightarrow \text{steering}.$$

This paper adds a controlled toy-world pillar, motivated by Othello-GPT world representations [11] and grokking via mechanistic interpretability [12], to test whether the proposed local-to-global objects detect known latent states before applying the same machinery to larger language-model decisions. The theory is explicitly local and piecewise: it does not assume that Transformer decision geometry is globally linear, but instead uses local linear and local quadratic response approximations inside fixed activation regimes while acknowledging globally nonlinear decision contours and superposed, non-orthogonal features.

1 Introduction

Modern language models do not make decisions in a single place. A factual answer, multiple-choice selection, next-token commitment, tool-use action, or reasoning branch emerges from distributed

computations across token positions, attention heads, MLPs, residual streams, normalization operations, and learned feature directions. Mechanistic interpretability has made this distribution visible: activation patching, causal tracing, attribution patching, circuit discovery, sparse autoencoders, and attribution graphs all show that behaviorally relevant information is spread across many internal sites rather than stored in one neuron or one scalar confidence value.

Yet our formal language for describing decisions often remains poorer than the computations we study. We typically ask which component is important, which feature is active, or how large the final logit margin is. These are useful questions, but they do not directly ask whether the model's distributed internal evidence coheres into a stable decision. A model may answer correctly because many local computations support the same conclusion. It may answer correctly but fragilely because a few late-layer features overpower unresolved conflict elsewhere. It may answer incorrectly because local evidence supporting different answers fails to integrate. It may hallucinate with high confidence because the final distribution is sharp even though intermediate computations remain mutually incompatible.

We propose a local-to-global view of Transformer decisions. For a fixed prompt and candidate answer contrast, we model internal evidence as a collection of local decision germs assigned to computational sites. These sites may be residual-stream vectors, attention-head outputs, MLP outputs, sparse-autoencoder features, or higher-level feature nodes. Edges between sites encode compatibility: how evidence at one site should transport to another if the model's computation is internally coherent with respect to the candidate contrast. A stable decision is then not identified with a literal single global neuron, nor with confidence alone, but with the degree to which these local germs can be glued into a compatible global decision structure.

The key object is an obstruction energy. In exact sheaf theory, one might be tempted to define an obstruction as a cohomology class $[ds] \in H^1$. That is not the right construction here: if s is a globally defined 0-cochain, then ds is a coboundary and its cohomology class is zero. Instead, we define obstruction metrically. Given a candidate-specific decision sheaf $\mathcal{F}_c$ over a computation complex K_x, and a local evidence state $s_c(x)$, the obstruction is the distance from $s_c(x)$ to the space of compatible global sections:

$$D_c(x) = \min_{g \in H^0(K_x; \mathcal{F}_c)} \|s_c(x) - g\|^2.$$

For computation, we use the equivalent finite-dimensional residual energy

$$O_c(x) = \|d_c s_c(x)\|^2 = \sum_{e:u \to v} w_e \left\|\rho^c_{u \to v} s_c(u) - s_c(v)\right\|^2.$$

This measures not whether the model is confident, but whether candidate-relevant internal evidence is compatible across sites.

Obstruction alone is not enough. A high obstruction score is scientifically useful only if it predicts causal interventions. We therefore couple obstruction to response theory. Given an activation patch, ablation, residual edit, steering vector, or feature intervention, we estimate its local effect on both obstruction and candidate margin. The resulting obstruction-response score ranks internal sites by predicted obstruction reduction, predicted margin change, and off-target KL/Fisher cost. The framework is falsifiable through a closed loop: compute obstruction, predict intervention sites, intervene, measure obstruction change, measure logit-margin change, and observe whether the model's decision flips, repairs, or stabilizes.

1.1 Professor-Feedback Addition: Representation Before Steering

The project should not be presented as beginning with steering. Steering presupposes that there is already a mathematically specified internal object to steer. The correct order is:

detect or represent what the model is doing $\longrightarrow$ monitor whether that representation is coherent $\longrightarrow$ steer or r

This changes the intellectual organization of the paper. The first task is not merely to find an intervention that changes an answer. The first task is to define a mathematical representation of a model's internal plan, feature, latent state, or task computation. In the present framework, this object is not a single activation direction. It is a collection of candidate-specific local germs, together with compatibility maps among them, over a computation complex.

This positioning also clarifies the relationship to standard mechanistic interpretability. A common feature-based workflow is:

find an activation direction $\longrightarrow$ find inputs that activate it $\longrightarrow$ interpret the direction as a concept.

That workflow is useful, but it is insufficient for the present goal because it does not define a distributed decision structure and does not, by itself, test whether local pieces of evidence are mutually compatible. Our framework treats feature detection as the first step, not the final step. Once a feature, plan, or latent state is represented, obstruction asks whether the distributed evidence for that state coheres, and response theory asks whether intervention on that structure changes behavior as predicted.

1.2 Contributions

This paper contributes:

(i) a local-to-global theory of Transformer decision formation;

(ii) a candidate-specific decision-sheaf formalism for internal evidence;

(iii) a corrected obstruction energy based on distance to global compatibility rather than the invalid cohomology class of an exact coboundary;

(iv) a causal response rule for predicting intervention-effective sites;

(v) an experimental protocol for testing repair, flip, and stabilization;

(vi) a limitation-aware account of non-identifiability, basis dependence, nonlinearity, and superposition.

The professor-feedback addition contributes an empirical and conceptual front-end to the theory:

(vii) controlled toy-world experiments where the latent state is known exactly;

(viii) explicit treatment of local linearity versus global nonlinearity;

(ix) explicit treatment of superposition and non-orthogonal feature decompositions;

(x) a representation-detection $\rightarrow$ monitoring $\rightarrow$ steering progression;

(xi) training-checkpoint experiments that track when mathematically crisp abstractions emerge.

2 The Paper's Goal and Thesis

The goal of this paper is to make local-to-global consistency a testable scientific theory of Transformer decision formation. In one sentence: we aim to show that candidate decisions in Transformers can be modeled as distributed compatibility events, and that a computable obstruction-response functional predicts causal interventions that repair, flip, or stabilize those decisions.

The central thesis is:

> **AI decisions are local-to-global consistency events: for a candidate decision, distributed local evidence across internal computational sites must become sufficiently compatible to support a stable global commitment.**

The reviewer-safe version is:

> For a fixed input and candidate contrast, a Transformer induces local decision evidence across computational sites. Stable decisions tend to exhibit low candidate-specific incompatibility relative to their output margin, while brittle, wrong, ambiguous, or overconfident decisions may exhibit measurable obstruction. A causal response rule can test this hypothesis by predicting which internal interventions should reduce obstruction and alter the final decision.

This is not a claim that all cognition is literally a global section. It is a modeling principle with a falsifiable instantiation.

2.1 Revised Thesis After Professor Feedback

The professor-feedback version of the thesis adds a prior requirement:

> a theory of steering must first be a theory of representation detection.

The revised thesis is:

> A model's decision should first be analyzed through the task-relevant internal representation or latent state it constructs. Local-to-global obstruction then measures whether the distributed evidence for that state and candidate decision is internally compatible. Causal response tests whether modifying that representation repairs, flips, or stabilizes behavior.

Thus the revised empirical program is:

controlled latent world → detect internal representation → measure compatibility → predict intervention → repai

3 Main Object: The Obstruction–Response Functional

The central object of the paper is the obstruction-response functional:

$$R_c(u, \delta; x) = \alpha\Big[- \Delta O_{c,u,\delta}(x) + \beta \Delta m_{c,u,\delta}(x) - \gamma \Delta \mathrm{Cost}_{u,\delta}(x)\Big].$$

Everything else serves this object. The computation complex defines where local evidence lives. The decision sheaf defines compatibility. The obstruction energy measures incompatibility. The response rule predicts how interventions change obstruction and margin. Decision boundaries explain flips. Geometry and Fisher penalties define intervention cost. Experiments test whether the functional predicts actual causal effects.

The professor-feedback addition does not replace this functional. It changes when the functional is introduced experimentally. Before applying R_c to steering, we first ask whether the model has a detectable internal representation of the task state. In toy worlds, this may be a board state, legal-move structure, modular sum, cyclic representation, parity variable, spatial relation, or object-permanence variable. Only after defining such a state do we define s_c, O_c, and R_c.

4 What This Paper Is and Is Not

This paper is	**This paper is not**
A theory of distributed decision formation	Just activation patching with new notation
A mathematical framework for local-to-global decision evidence	Just sheaf theory applied decoratively
A causal intervention method for repairing, flipping, and stabilizing decisions	Just topology, QFT, or differential geometry
An empirical test of local-to-global consistency in Transformers	Just a hallucination detector
A framework for measuring incompatibility beyond confidence and margin	A benchmark-only paper
A foundation for future AI decision science	A claim that all AI cognition is literally a global section
A limitation-aware method for intervention-effective structures	A claim to identify the unique true circuit
A framework that begins with representation detection before steering	A claim that a single activation direction uniquely equals a concept
A local and piecewise response theory for globally nonlinear networks	A globally linear theory of Transformer computation
A causal test of mathematically defined latent states in toy worlds	A claim that toy worlds prove all LLM cognition works the same way

5 Formal Framework

5.1 Candidate Contrasts

Let M_θ be a frozen Transformer and x an input prompt. For candidate answers a and b, define the candidate contrast

$$c = (a, b).$$

The candidate margin is

$$m_c(x) = z_a(x) - z_b(x),$$

where z_a, z_b are logits or task scores.

In controlled toy worlds, a and b may be legal versus illegal moves, correct versus incorrect board-state predictions, correct versus incorrect modular sums, or valid versus invalid actions. This makes the candidate contrast semantically grounded rather than only logit-defined.

5.2 Computation Complex

Definition 1 (Computation complex)**.** *For an input x, the computation complex is a finite directed complex*

$$K_x = (V_x, E_x),$$

where vertices $v \in V_x$ are internal computational sites and edges $e : u \to v$ encode computational, causal, or learned compatibility relations.

A vertex may be indexed as

$$v = (\ell, t, q),$$

where ℓ is layer, t is token position, and q is a component type such as residual stream, attention-head output, MLP output, sparse-autoencoder feature block, or transcoder feature.

For toy-world experiments, q may also refer to a probe or feature block associated with an externally known latent variable, such as board occupancy, piece color, move legality, modular phase, parity, order, or spatial adjacency. This is not an extra assumption about large language models; it is a controlled validation setting where the latent state is known.

5.3 Local Decision Germs

Definition 2 (Candidate-specific stalk)**.** *For a candidate contrast c, each site $v \in V_x$ is assigned a finite-dimensional evidence space*

$$\mathcal{F}_c(v) \cong \mathbb{R}^{r_v}.$$

Let $\phi_v(x) \in \mathbb{R}^{d_v}$ be the activation or feature vector at site v. A local decision germ is

$$s_c(v; x) = P_v^c \phi_v(x) \in \mathcal{F}_c(v),$$

where P_v^c is a candidate-specific projection, probe, feature map, or logit-lens contrast. The full local evidence state is a 0-cochain:

$$s_c(x) \in C^0(K_x; \mathcal{F}_c) = \prod_{v \in V_x} \mathcal{F}_c(v).$$

5.4 Task-Relevant Latent-State Germs

The professor-feedback addition requires a more general object before steering: a task-relevant latent-state germ. Suppose a controlled task has a latent state

$$y(x) \in \mathcal{Y},$$

such as an Othello board state, a set of legal moves, or a modular sum. A site-level latent-state germ is

$$s_y(v; x) = P_v^y \phi_v(x),$$

where P_v^y is trained or defined to extract the part of $\phi_v(x)$ relevant to y. Candidate-specific decision germs are then a special case in which the latent state is projected onto evidence for a candidate contrast:

$$s_c(v;x) = Q_v^c s_y(v;x).$$

This makes explicit that steering should not be attempted until the relevant internal representation has been defined and tested.

5.5 Compatibility Maps

For every edge $e : u \to v$, define a candidate-specific transport map

$$\rho_{u\to v}^c : \mathcal{F}_c(u) \to \mathcal{F}_c(v).$$

In experiments, these maps may be learned by ridge regression on clean examples:

$$\rho_{u\to v}^c = \arg\min_R \sum_{x_i \in T_{\text{clean}}} \|R s_c(u;x_i) - s_c(v;x_i)\|^2 + \lambda \|R\|_F^2.$$

They may also be approximated through local Jacobians:

$$\rho_{u\to v}^c \approx P_v^c \frac{\partial \phi_v}{\partial \phi_u} (P_u^c)^\dagger.$$

In toy-world representation experiments, compatibility maps can be trained not only between candidate evidence spaces but also between latent-state evidence spaces:

$$\rho_{u\to v}^y : \mathcal{F}_y(u) \to \mathcal{F}_y(v).$$

For example, in an Othello-style task, compatibility can require that board-state evidence at an early site transports consistently to legality evidence at a later site. In modular addition, compatibility can require that local Fourier-like phase evidence transports consistently toward the predicted modular sum.

5.6 Coboundary and Global Sections

Define the coboundary operator

$$d_c : C^0(K_x; \mathcal{F}_c) \to C^1(K_x; \mathcal{F}_c)$$

by

$$(d_c s_c)(e : u \to v) = \rho_{u\to v}^c s_c(u) - s_c(v).$$

The compatible global decision sections are

$$H^0(K_x; \mathcal{F}_c) = \ker d_c = \{g \in C^0(K_x; \mathcal{F}_c) : d_c g = 0\}.$$

5.7 Corrected Obstruction Energy

A tempting but incorrect construction is to call $[d_c s_c] \in H^1(K_x; \mathcal{F}_c)$ the obstruction. This is invalid in the present setting because s_c is a globally defined 0-cochain, so $d_c s_c$ is exact and its cohomology class is zero. Instead, the obstruction is metric.

Definition 3 (Distance-to-global-section obstruction)**.** *The obstruction of $s_c(x)$ is*

$$D_c(x) = \min_{g \in H^0(K_x; \mathcal{F}_c)} \|s_c(x) - g\|_M^2.$$

Definition 4 (Laplacian residual obstruction)**.** *The computable obstruction energy is*

$$O_c(x) = \|d_c s_c(x)\|_W^2 = \sum_{e:u \to v} w_e \left\|\rho^c_{u \to v} s_c(u; x) - s_c(v; x)\right\|^2.$$

Let

$$L_c = d_c^* W d_c.$$

Then

$$O_c(x) = s_c(x)^\top L_c s_c(x).$$

Normalize obstruction by evidence magnitude:

$$\widetilde{O}_c(x) = \frac{O_c(x)}{\|s_c(x)\|_M^2 + \epsilon}.$$

Define fragility by comparing obstruction to margin:

$$\mathrm{Frag}_c(x) = \frac{\widetilde{O}_c(x)}{|m_c(x)| + \epsilon}.$$

5.8 Latent-State Obstruction

For controlled latent-state tasks, define an analogous latent-state obstruction

$$O_y(x) = \sum_{e:u \to v} w_e \left\|\rho^y_{u \to v} s_y(u; x) - s_y(v; x)\right\|^2.$$

This measures incompatibility among local representations of the task state itself, not merely incompatibility among final-answer evidence. In a board game, O_y can measure inconsistency among local board-state, occupancy, color, and legality representations. In modular arithmetic, O_y can measure inconsistency among local cyclic or Fourier-like representations of the modular structure.

6 Obstruction–Response Intervention Theory

Let h_u be the activation at site u, and let an intervention δ_u be injected by

$$h \mapsto h + B_u \delta_u.$$

The linearized margin response is

$$\Delta m_c \approx J^m_{c,u}\delta_u, \qquad J^m_{c,u} = \frac{\partial m_c}{\partial h_u}.$$

The obstruction response is

$$\Delta O_c \approx \nabla_{h_u} O_c^\top \delta_u + \frac{1}{2}\delta_u^\top H_u^O \delta_u.$$

To penalize collateral damage, define an off-target KL or Fisher cost:

$$\Delta\text{Cost}_u \approx \frac{1}{2}\delta_u^\top F_u \delta_u,$$

where

$$F_u = \mathbb{E}_{y\sim p_\theta(\cdot|x)}\left[\nabla_{h_u}\log p_\theta(y|x)\nabla_{h_u}\log p_\theta(y|x)^\top\right].$$

The intervention objective is

$$\delta_u^\star = \arg\min_\delta \left[O_c(h + B_u\delta) - \beta m_c(h + B_u\delta) + \frac{\mu}{2}\delta^\top F_u\delta + \frac{\lambda}{2}\|\delta\|^2\right].$$

Under a local quadratic approximation,

$$\delta_u^\star = -\left(H_u^O + \mu F_u + \lambda I\right)^{-1}\left(\nabla_{h_u} O_c - \beta(J^m_{c,u})^\top\right).$$

For practical ranking, define

$$S_u = \alpha\left[-\Delta O_{c,u} + \beta\Delta m_{c,u} - \gamma\Delta\text{Cost}_u\right].$$

6.1 Standardized Response Score

Empirical component-level tests can reveal scale pathologies: one term in S_u may numerically dominate the others even when it does not add causal predictive value. Therefore a standardized ranking score should also be tested:

$$S^z_u = z_u(\Delta m_c) - \alpha z_u(\Delta O_c) - \gamma z_u(\text{Cost}),$$

where $z_u(\cdot)$ denotes standardization across candidate sites for the same example. This score is not a new theory. It is a necessary empirical control ensuring that obstruction is tested as an additive signal rather than as a scale artifact. The obstruction term survives only if it improves out-of-sample intervention prediction beyond attribution-only, activation norm, cost, and gradient saliency baselines.

7 Local Linearity, Piecewise Geometry, and Nonlinearity

A central mathematical issue raised by the professor is nonlinearity. The framework must not pretend that a Transformer is globally linear. Attention patterns, layer normalization, MLP nonlinearities, softmax operations, and residual interactions can produce highly nonlinear decision geometry. The appropriate claim is local and piecewise: around a fixed input, candidate contrast,

and intervention scale, response can often be approximated by a local linear or local quadratic model.

A piecewise-linear analogy is useful. A globally nonlinear decision contour may be segmented into local regions in which the active computation is approximately affine. Inside one such region,

$$m_c(h) \approx A_c h + b_c,$$

and

$$O_c(h+\delta) \approx O_c(h) + \nabla_h O_c^\top \delta + \frac{1}{2}\delta^\top H^O \delta.$$

This does not assert that the global hypercontour $\{h : m_c(h) = 0\}$ is linear. It asserts that response predictions are local approximations whose validity must be empirically tested.

The paper should therefore report where local response succeeds and where it fails. Failure is expected when:

1. small interventions change active attention pathways;
2. layer normalization induces strong nonlocal effects;
3. the response map loses rank;
4. the model crosses between piecewise regions;
5. the obstruction projection becomes ill-conditioned;
6. the intervention is too large for the local approximation.

This treatment directly addresses the professor's point: if nonlinearities are not modeled globally, the paper must still explain why local approximations are scientifically meaningful and what regime they apply in.

8 Superposition, Non-Orthogonality, and Non-Identifiability

A second mathematical issue raised by the professor is superposition. A model with hidden dimension d can represent more than d human-interpretable features because features need not occupy mutually orthogonal directions. They may be non-orthogonal, distributed, and superposed. Therefore neither spectral analysis of a weight matrix nor interpretation of a single activation direction can uniquely recover the semantic feature set.

Let $h \in \mathbb{R}^d$ be an activation. In a superposed representation, one may have multiple decompositions

$$h \approx \sum_{i \in I} a_i f_i \approx \sum_{j \in J} b_j g_j,$$

where the feature families $\{f_i\}$ and $\{g_j\}$ are non-orthogonal and semantically different. This makes the statement "the model is thinking about concept X because the activation points in direction v" non-unique.

The obstruction-response framework should not claim to find the unique true feature basis. Its safer claim is:

> we identify candidate-specific compatibility structures that are intervention-effective.

That is, a structure is scientifically useful if it:

1. predicts behavior;
2. predicts causal interventions;
3. remains stable across examples or checkpoints;
4. beats random, norm, attribution-only, and generic-projection baselines;
5. survives out-of-sample validation.

Superposition is therefore not a side limitation. It is a motivation for the framework. When decomposition into individual features is non-unique, the object of study should be a causal compatibility structure rather than a single semantic direction.

9 Algorithm

Algorithm 1 Obstruction–Response Intervention for Transformer Decisions

Require: Model M_θ, prompt x, candidates a, b, sites V, edges E, learned maps ρ_e^c, intervention budget k

Ensure: Obstruction score, fragility score, ranked intervention sites, predicted and observed decision changes

Define candidate contrast $c = (a, b)$
Run $M_\theta(x)$ and cache activations $\{\phi_v(x)\}_{v \in V}$
Compute local germs $s_c(v; x) = P_v^c \phi_v(x)$
Compute margin $m_c(x) = z_a(x) - z_b(x)$
for each edge $e : u \to v$ **do**
    Compute residual $r_e = \rho_e^c s_c(u; x) - s_c(v; x)$
end for
Compute obstruction $O_c(x) = \sum_e w_e \|r_e\|^2$
Compute normalized obstruction $\widetilde{O}_c(x)$ and fragility $\mathrm{Frag}_c(x)$
for each intervention site u **do**
    Choose intervention direction δ_u by patching, ablation, residual edit, or feature steering
    Estimate $\Delta m_{c,u} = J_{c,u}^m \delta_u$
    Estimate $\Delta O_{c,u} = \nabla_{h_u} O_c^\top \delta_u + \frac{1}{2} \delta_u^\top H_u^O \delta_u$
    Estimate off-target cost $\Delta\mathrm{Cost}_u$
    Score $S_u = \alpha[-\Delta O_{c,u} + \beta \Delta m_{c,u} - \gamma \Delta\mathrm{Cost}_u]$
end for
Patch, ablate, or steer the top k sites by S_u
Re-run model and measure actual $\Delta O_c, \Delta m_c, \mathrm{KL}$, and decision outcome
return scores, rankings, predicted changes, observed changes

Algorithm 2 Professor-Feedback Pipeline: Representation Detection Before Steering

Require: Controlled task with latent state $y(x)$, model checkpoints θ_t, sites V, edges E
Ensure: Detected representation, monitoring signal, obstruction score, causal intervention result

1: Train or load a small Transformer on a controlled task
2: For each input x, compute the true latent state $y(x)$ from the task simulator
3: Cache activations $\{\phi_v(x)\}_{v\in V}$
4: Fit or define latent-state maps P_v^y and compute $s_y(v;x) = P_v^y \phi_v(x)$
5: Test whether s_y decodes the true latent state out of sample
6: Fit compatibility maps $\rho_{u\to v}^y$ on clean examples
7: Compute latent-state obstruction $O_y(x)$
8: Identify failures where output is wrong, illegal, or unstable
9: Distinguish whether failure comes from wrong latent state, wrong computation from latent state, or wrong output policy
10: Intervene on the latent representation $h_u \mapsto h_u + \delta_y$
11: Measure whether the intervention reduces O_y, changes m_c, and repairs behavior
12: Across checkpoints, track abstraction strength $A_y(t)$, obstruction $O_y(t)$, and task accuracy

10 Decision Boundaries, Flips, and Singular Regimes

A candidate decision flips when an intervention moves the model state across the boundary

$$m_c(h) = 0.$$

Under local linearization, frozen attention, or a fixed piecewise-linear activation region, the margin is approximately affine:

$$m_c(h) \approx A_c h + b_c.$$

Thus, locally, the decision boundary is

$$\{h : A_c h + b_c = 0\}.$$

Singular regimes arise when the response map loses rank, when the projection onto $H^0(K_x; \mathcal{F}_c)$ becomes ill-conditioned, or when small interventions cause discontinuous changes in active computational pathways. Algebraic, tropical, or singularity-theoretic language should be used only under such explicit approximations.

The professor-feedback addition makes this section more central. The correct statement is not that Transformer decision boundaries are globally affine. The correct statement is that within a fixed local computational regime, the boundary can be approximated by an affine hyperplane. Globally, the decision surface may be a deeply nonlinear hypercontour assembled from many local regions. Therefore all response claims must report the intervention scale and the diagnostic regime in which local approximation was tested.

11 Theorem Targets

Theorem 1 (Energy–distance equivalence)**.** *Let $d_c : C^0 \to C^1$ be the coboundary of a finite Hilbert sheaf with positive edge weights W, and let $L_c = d_c^* W d_c$. Let $P_{\ker d_c}$ be the M-orthogonal projection onto $H^0 = \ker d_c$. Then, on $(\ker d_c)^\perp$, there exist spectral constants $0 < \lambda_{\min}^+ \le \lambda_{\max}$ such that*

$$\lambda_{\min}^+ \|(I - P_{\ker d_c})s\|_M^2 \le s^\top L_c s \le \lambda_{\max} \|(I - P_{\ker d_c})s\|_M^2.$$

Proposition 1 (Minimum-energy repair under linear response)**.** *Assume O_c is locally quadratic, m_c is locally linear, and the off-target cost is quadratic with positive semidefinite Fisher matrix F_u. Then the local optimal repair intervention is*

$$\delta_u^\star = -\left(H_u^O + \mu F_u + \lambda I\right)^{-1}\left(\nabla_{h_u} O_c - \beta (J_{c,u}^m)^\top\right).$$

Proposition 2 (Obstruction–margin stability)**.** *Suppose margin response is locally Lipschitz under perturbations of norm at most r. If*

$$|m_c(x)| > \sup_{\|\delta\| \le r} |J^m \delta| + \tau O_c(x),$$

then the candidate ordering is stable under the allowed perturbation class.

Proposition 3 (Non-identifiability)**.** *In redundant or superposed representations, multiple low-overlap compatibility structures may induce similar obstruction-response predictions for the same candidate contrast.*

Proposition 4 (Local decision boundary)**.** *Under frozen attention and a fixed piecewise-linear activation region, a candidate margin boundary is locally affine:*

$$\{h : m_c(h) = 0\} = \{h : A_c h + b_c = 0\}.$$

Proposition 5 (Relative extension obstruction)**.** *Let $U \subset K$ be a subcomplex and let s_U be a fixed partial section. A true cohomological obstruction to extending s_U to a global section can live in relative cohomology:*

$$\mathrm{Obs}(s_U) \in H^1(K, U; \mathcal{F}).$$

Proposition 6 (Local validity of response approximation)**.** *Let $R(h)$ denote either $m_c(h)$ or $O_c(h)$. Suppose R is twice differentiable in a neighborhood $B_r(h_0)$ of the activation state h_0. Then for interventions δ with $\|\delta\| \le r$,*

$$R(h_0 + \delta) = R(h_0) + \nabla R(h_0)^\top \delta + \frac{1}{2}\delta^\top H_R(h_0)\delta + o(\|\delta\|^2).$$

The obstruction-response approximation is valid only to the extent that interventions remain inside such a local region.

Proposition 7 (Superposition non-uniqueness)**.** *Let a representation space contain non-orthogonal feature families $\{f_i\}$ and $\{g_j\}$. If an activation h admits low-error decompositions in both families, then feature attribution is not unique. Therefore the framework cannot claim to identify a unique semantic decomposition without additional constraints. The empirically testable claim is instead intervention effectiveness of a candidate-specific compatibility structure.*

12 Experimental Blueprint

The experiments should tell a cumulative story rather than merely demonstrate one trick.

12.1 Experiment 0: Controlled Toy Worlds for Representation Detection

The professor-feedback addition inserts a new first experiment before IOI. The purpose is to test the framework in a setting where the latent state is known exactly. This prevents the paper from jumping directly to steering without first validating the representation being steered.

The controlled progression is:

known latent state → internal representation → compatibility/obstruction → causal intervention → behavioral re

12.1.1 Experiment 0A: Othello-Style World-State Representations

This experiment follows Li et al.'s Othello-GPT setting [11]. A GPT-style model is trained only on Othello move sequences and learns to predict legal moves. The model is not explicitly given the board state. Nevertheless, Li et al. find evidence for an emergent nonlinear internal representation of the board state and show that interventions on this representation can affect the model's move predictions.

Let $x_{1:t}$ be a sequence of Othello moves and let

$$B_t \in \mathcal{B}$$

be the true board state after these moves. A simulator provides:

$$B_t, \qquad \mathcal{L}(B_t), \qquad \mathcal{I}(B_t),$$

where $\mathcal{L}(B_t)$ is the set of legal moves and $\mathcal{I}(B_t)$ is the set of illegal moves.

The representational question is:

$$\text{Does the model internally encode } B_t?$$

For each site v, define a board-state germ

$$s_B(v; x) = P_v^B \phi_v(x).$$

For a candidate move contrast $c = (a, b)$, where $a \in \mathcal{L}(B_t)$ and $b \in \mathcal{I}(B_t)$, define a decision germ

$$s_c(v; x) = Q_v^c s_B(v; x).$$

The obstruction energy becomes

$$O_c(x) = \sum_{e:u\to v} w_e \left\| \rho_{u\to v}^c s_c(u; x) - s_c(v; x) \right\|^2.$$

In this setting, obstruction has a concrete interpretation: incompatibility between local board-state evidence, local legality evidence, and local move evidence.

The experiment separates three failure modes:

1. **World-state failure:** the decoded board state $\widehat{B}_t$ is wrong.
2. **Legality-computation failure:** the board state is decoded correctly, but the inferred legal moves are wrong.
3. **Policy failure:** the board and legality are represented correctly, but the model assigns high probability to an illegal or bad move.

The key causal test is:

correct internal board or legality representation $\implies -\Delta O_c > 0 \implies \Delta m_c > 0 \implies$ illegal move repaired.

Success criterion. Obstruction-response succeeds only if it predicts interventions that repair illegal moves better than random, activation norm, gradient saliency, attribution patching, and patching heuristics. It is not enough for a probe to decode the board state; the representation must be causally involved in the output.

12.1.2 Experiment 0B: Modular Addition and Grokking

This experiment follows Nanda et al.'s grokking work on modular addition [12]. A small Transformer is trained on

$$a + b \pmod{p}.$$

Checkpoints are saved throughout training. The known abstraction is cyclic structure, and Nanda et al. show that mechanistic progress measures can reveal phases such as memorization, circuit formation, and cleanup.

Let θ_t be the model parameters at training step t. Define an abstraction-strength measure

$$A_{\text{cyc}}(t) = \text{degree to which the model internally represents modular cyclicity.}$$

This may be estimated through Fourier concentration, circular geometry in activation space, causal ablation of cyclic subspaces, or compatibility among modular-evidence germs.

For candidate contrast $c = (a, b)$, where a is the correct modular sum and b is an incorrect sum, track

$$t \mapsto O_c(x; \theta_t), \qquad t \mapsto A_{\text{cyc}}(t), \qquad t \mapsto \text{Acc}_{\text{test}}(t).$$

The central hypothesis is:

$$A_{\text{cyc}}(t) \uparrow \implies O_c(x; \theta_t) \downarrow \implies m_c(x; \theta_t) \uparrow .$$

This tests whether representational emergence precedes behavioral emergence.

Success criterion. The experiment supports the paper if obstruction or compatibility measures track formation of the generalizing algorithm before test accuracy fully improves, and if ablation of the detected cyclic subspace destroys generalization while preserving some memorized behavior.

12.2 Experiment 1: IOI and Known Circuits

Question. Does obstruction-response predict which residual-stream or component sites causally repair corrupted IOI decisions?

Models and tasks. Use GPT-2 small with TransformerLens on IOI clean/corrupt prompts.

Metrics. Top-k repair rate, predicted-vs-observed Δm_c, predicted-vs-observed ΔO_c, AUC for effective sites, and off-target KL.

Supporting result. Obstruction-response rankings outperform random, activation norm, gradient saliency, attribution patching, and simple patching heuristics.

Weakening result. The score is no better than attribution patching, or obstruction does not change when patches repair the answer.

Current empirical status. The initial IOI implementation should be treated as calibration, not as the main positive result. Scalar residual-chain obstruction, multi-token scalar obstruction, naive low-rank residual obstruction, residualized low-rank obstruction, and whitened low-rank obstruction did not yield a clean obstruction-specific signal. Component-level attribution patching

did produce strong causal signal, but current obstruction-response scores did not cleanly beat attribution-only. Therefore IOI should remain an attribution-positive and obstruction-stress-test section unless standardized obstruction-response validation shows out-of-sample lift over attribution.

12.3 Experiment 2: Factual Recall

Question. Does obstruction-response generalize from synthetic circuits to factual decisions?

Models and tasks. Use GPT-2 medium, Pythia models, and factual recall datasets such as CounterFact or zsRE-style prompts.

Metrics. Repair rate for corrupted or counterfactual prompts, margin change, obstruction reduction, and KL collateral damage.

Supporting result. Predicted interventions repair factual answers with fewer sites and lower KL than baselines.

12.4 Experiment 3: Multiple-Choice and Reasoning Decisions

Question. Does the framework apply to candidate decisions beyond next-token factual recall?

Models and tasks. Use ARC, MMLU subsets, or HellaSwag with candidate contrasts $c = (\text{correct}, \text{distractor})$.

Metrics. Candidate-margin improvement, answer flips, calibration of flip probability, and transfer across paraphrases.

12.5 Experiment 4: Ambiguity and Conflicting Evidence

Question. Do ambiguous prompts produce high obstruction relative to margin?

Models and tasks. Construct prompts with misleading context, conflicting cues, or late corrections.

Metrics. Fragility, paraphrase instability, repair by targeted intervention, and obstruction change after disambiguating context.

12.6 Experiment 5: Hallucination and Overconfidence

Question. Can obstruction identify overconfident but internally incompatible decisions?

Models and tasks. Use TruthfulQA-style prompts or factual prompts with uncertain premises.

Metrics. Obstruction at fixed confidence, semantic-entropy comparison, and intervention sensitivity.

The toy-world illegal-move setting provides the cleanest operational analogue of hallucination: the model predicts an action inconsistent with the true latent world state. This should be used to validate the distinction between wrong internal state, wrong computation from state, and wrong output policy before making large-language-model hallucination claims.

12.7 Experiment 6: SAE-Feature Version

Question. Does feature-space obstruction improve over residual-space obstruction?

Models and tasks. Use open-weight models with available sparse autoencoders.

Metrics. Intervention efficiency, interpretability of selected sites, and robustness to feature basis.

13 Baselines and Ablations

13.1 Baselines

The minimal strong baseline set is:

- random intervention sites;
- activation norm;
- gradient saliency;
- logit margin and entropy;
- attribution patching;
- activation patching sweep as an oracle;
- path patching where feasible;
- causal tracing for factual recall;
- ACDC or edge-attribution patching where feasible;
- SAE feature magnitude and SAE steering in feature experiments;
- Fisher norm;
- intrinsic dimension and persistent homology as diagnostic-only baselines;
- EICS-style sheaf inconsistency score where implementation is feasible.

Additional baselines required by the professor-feedback additions:

- linear probe accuracy without causal intervention;
- causal intervention on decoded latent state;
- random latent-state subspace;
- generic PCA subspace;
- spectral directions of weights or activations;
- attribution-only standardized score;
- attribution plus cost but no obstruction;
- attribution plus activation norm but no obstruction;
- checkpoint-wise behavioral accuracy versus checkpoint-wise representation strength.

13.2 Ablations

Essential ablations:

- no candidate-specific sheaf;
- identity transports;
- random transports;
- learned transports versus Jacobian transports;
- scalar logit-lens germs versus low-rank residual germs;
- residual features versus SAE features;
- obstruction without response;
- response without obstruction;
- no Fisher/KL penalty;
- different edge weights;
- different site granularities;
- paraphrase transfer;
- cross-model transfer.

Professor-feedback ablations:

- local-linear approximation versus measured nonlinear response;
- small versus large intervention radius;
- fixed attention pattern versus recomputed attention pattern;
- orthogonal spectral basis versus non-orthogonal learned feature basis;
- probe-only representation versus causally validated representation;
- latent-state obstruction versus candidate-margin obstruction;
- pre-grokking, grokking, and post-grokking checkpoints;
- representation emergence before versus after behavioral emergence.

14 Expected Figures and Tables

- Figure 1: Local-to-global decision schematic. A Transformer computation graph with local evidence at tokens, layers, heads, MLPs, residual streams, and features. Compatible evidence glues into stable commitment; incompatible evidence produces obstruction.
- Figure 2: Mathematical object diagram. Show K_x, stalks $\mathcal{F}_c(v)$, germs $s_c(v)$, maps $\rho^c_{u\to v}$, residuals r_e, global section space H^0, and obstruction distance.
- Figure 3: Obstruction-response loop:

$$\text{local evidence} \to O_c \to S_u \to \text{intervention} \to \Delta O_c \to \Delta m_c \to \text{repair/flip/stabilize}.$$

- Figure 4: Predicted versus observed margin change. A scatter plot across sites, prompts, and tasks showing $\widehat{\Delta m_c}$ or S_u versus observed Δm_c.
- Figure 5: Obstruction reduction versus repair probability. Bucket interventions by actual $-\Delta O_c$ and show monotonic repair probability.
- Figure 6: Baseline comparison. Top-k repair rate or AUC against attribution patching, activation norm, gradient saliency, random, and oracle activation patching.
- Figure 7: Failure cases. Examples where obstruction decreases without margin movement, margin moves without obstruction reduction, or transport maps fail.

Additional figures from professor feedback:

- Figure A: Toy-world pipeline:

  game transcript → hidden board state → internal representation → obstruction → intervention → legal-mov

- Figure B: Emergence over training. Plot

$$A_{\text{cyc}}(t), \qquad O_c(t), \qquad \text{Acc}_{\text{test}}(t)$$

  over checkpoints.

- Figure C: Failure-mode separation:

  wrong latent state, wrong legality computation, wrong output policy.

- Figure D: Local versus global geometry. Show a globally nonlinear decision contour segmented into local approximately affine regions.
- Figure E: Superposition schematic. Show non-orthogonal features producing non-unique decompositions of the same activation vector.

15 Hierarchy of Ideas

Idea	Role
Local-to-global decision theory	Central spine
Obstruction-response functional	Central spine and main object
Sheaf/compatibility structure	Major mathematical section
Corrected obstruction energy	Major mathematical section
Causal response	Major algorithmic/intervention section
Activation patching	Empirical intervention tool and baseline
Causal abstraction	Related work and conceptual support
Sparse autoencoders	Major optional feature-space experiment
Information geometry	Subsection for Fisher/KL intervention cost
Decision singularities	Subsection/appendix under local boundary assumptions
Field theory	Supporting response-calculus language only
Topology	Optional diagnostic/appendix
Algebraic/tropical geometry	Appendix only under frozen-attention or local-linear assumptions
Stochastic thermodynamics	Future work unless directly tested
Hallucination	Failure-mode experiment, not central theorem
Theorem proving	Future extension or brief experiment
Agents/search	Future paper material
Renormalization/coarse-graining	Future extension unless used for site compression
Representation detection	Necessary stage before monitoring and steering
Controlled toy worlds	First empirical pillar for known latent states
Nonlinearity and piecewise geometry	Required mathematical story for local approximations
Superposition and non-orthogonal features	Motivation for compatibility structures rather than single directions
Grokking/checkpoint dynamics	Experiment on emergence of mathematical abstractions

16 Title Options

1. Causal Obstruction–Response Theory of Transformer Decisions
2. Local-to-Global Decision Mechanics in Transformers
3. Obstruction-Guided Decision Mechanics in AI Systems
4. Decision Coherence in Transformers: A Local-to-Global Theory
5. Local Evidence, Global Commitment: A Theory of Transformer Decisions
6. Obstruction, Response, and Repair in Transformer Decision-Making

7. Causal Decision Sheaves for Transformer Interpretability
8. Measuring and Repairing Local-to-Global Inconsistency in Language Models
9. The Mechanics of Transformer Decisions
10. From Local Evidence to Global Commitment in AI Systems
11. Decision Singularity Surgery in Transformers
12. A Local-to-Global Calculus for AI Decisions
13. Obstruction Energy Predicts Causal Decision Repair in Transformers
14. Transformer Decisions as Distributed Consistency Events
15. Toward a Science of AI Decision Mechanics

The recommended title is:

Causal Obstruction–Response Theory of Transformer Decisions

17 Novelty and Safety

17.1 Unsafe Claims

Avoid:

- "We introduce the first use of sheaf theory in Transformers."
- "We introduce the first local-to-global theory of neural networks."
- "Hallucination is obstruction."
- "We prove AI decisions are global sections."
- "Field theory of Transformer patching is new."
- "We identify the true circuit."
- "Wrong answers are exactly failures of gluing."
- "This is a universal theory of intelligence."
- "A single activation direction is the model's concept."
- "The model is globally linear."
- "Toy-world success proves LLM hallucination is solved."
- "Spectral directions recover the true feature basis."

17.2 Safe Claims

Use:

- "We propose a candidate-specific local-to-global framework for Transformer decisions."
- "We define obstruction as distance to global compatibility, computed by a Laplacian residual."
- "We test whether obstruction-response rankings predict causal intervention effects."
- "The method identifies intervention-effective structures, not necessarily unique circuits."
- "Obstruction complements confidence, entropy, and margin by measuring internal incompatibility."
- "Local-to-global consistency is a modeling principle, not a universal law."
- "The framework begins by detecting task-relevant latent representations before steering them."
- "The response theory is local and must be tested against nonlinear effects."
- "Controlled toy worlds provide known latent states for validating representation detection, monitoring, and intervention."
- "Superposition motivates causal compatibility tests rather than unique semantic decomposition claims."

18 Reviewer-Proof Positioning

1. **This is just activation patching with extra math.** Activation patching measures effects by intervention; obstruction-response predicts which interventions should work and why.
2. **The sheaf is arbitrary.** The paper includes identity, random, learned, and Jacobian transport ablations.
3. **Transport maps are post hoc.** Transports are trained on clean examples and tested out-of-distribution on corrupt, ambiguous, and paraphrased examples.
4. **Obstruction only tracks margin.** Report partial correlations and predictive lift over margin and entropy.
5. **This does not prove decisions are global sections.** The paper does not claim literal global-section cognition; it tests a compatibility model.
6. **SAE bases are unstable.** Compare residual, PCA, probe, and SAE bases.
7. **Universality is overclaimed.** Universal language is restricted to discussion; main claims are empirical and candidate-specific.
8. **Field theory and topology are decorative.** They are not central; response theory and obstruction are central.
9. **Related work already uses sheaves.** The novelty is candidate-specific obstruction-response and causal intervention prediction.

10. **Interventions do not transfer.** Transfer is explicitly measured; lack of transfer becomes a limitation.

11. **Attribution patching is stronger.** Attribution patching is a primary baseline.

12. **Patching conclusions depend on metric choice.** The paper reports multiple metrics: margin, KL, repair rate, flip probability, and obstruction change.

13. **Non-unique circuits undermine the framework.** The framework claims intervention-effective structures, not unique mechanisms.

14. **Obstruction reduction may be epiphenomenal.** The paper tests mediation by measuring whether $-\Delta O_c$ predicts Δm_c after controls.

15. **Math is too heavy.** Only the obstruction definition, energy-distance theorem, and response rule are central.

Additional reviewer objections introduced by the professor feedback:

16. **You are steering before defining what is being steered.** The revised paper begins with representation detection and controlled latent-state tasks.

17. **Your method ignores nonlinearities.** The theory is explicitly local and piecewise; global nonlinear decision contours are acknowledged, and local approximation validity is empirically tested.

18. **Feature directions are non-unique because of superposition.** The paper does not claim unique semantic decomposition. It tests intervention-effective compatibility structures.

19. **Spectral matrix analysis could already find the features.** Orthogonal spectral directions are insufficient in superposed representations; our framework allows non-orthogonal, candidate-specific, distributed structures.

20. **Toy models are too small.** Toy worlds are not used to prove full LLM cognition. They are used to validate the framework when the true latent state and failure mechanism are known exactly.

19 Minimum Viable, Strong, and Landmark Versions

19.1 Minimum Viable Paper

Scope. GPT-2 small and Pythia-small models; IOI and one factual recall task.

Theorem set. Energy-distance equivalence, minimum-energy repair, and cohomology correction.

Claims. Obstruction-response predicts causal repair sites better than random, activation norm, gradient saliency, and basic attribution patching on at least one known-circuit task.

Revised minimum viable paper. Add one controlled toy-world experiment before IOI. The minimum viable version should show that in a known latent-state task, such as Othello-style legal move prediction or modular addition, the method detects an internal representation, monitors its coherence, and causally intervenes on it.

19.2 Strong Conference Paper

Scope. GPT-2, Pythia, and one modern open-weight model; IOI, factual recall, MCQA, and ambiguity.

Theorem set. Energy-distance equivalence, repair rule, obstruction-margin stability, non-identifiability, local boundary theorem in appendix.

Claims. Obstruction-response generalizes across tasks and adds predictive value beyond strong patching and saliency baselines.

Revised strong conference paper. Add both controlled toy-world pillars: Othello-style world-state representation and modular-addition grokking. Show that abstraction strength and obstruction can be tracked over training checkpoints.

19.3 Landmark Paper

Scope. Multiple models, multiple tasks, residual and feature-space implementations, robust transfer, public benchmark.

Theorem set. All main results plus empirically validated linear-response regimes.

Claims. Local-to-global obstruction is a causal predictor of decision repair and stability, providing a new way to study AI decision mechanics.

The unforgettable figure would show predicted obstruction-response score versus observed causal margin change across tasks and models, with clear superiority over attribution patching and other baselines.

Revised landmark paper. The landmark claim is not merely that obstruction improves patching. It is that mathematically crisp abstractions can emerge inside vector-space models, can be detected as distributed compatibility structures, can be monitored during inference and training, and can be causally manipulated to repair behavior.

20 Updated Roadmap

The revised roadmap is:

1. **Toy Othello / board-game model.** Train or use an Othello-GPT-style model. Decode board state. Measure obstruction between board evidence, legality evidence, and move evidence. Intervene on board-state features and test legal-move repair.

2. **Grokking / modular arithmetic model.** Train a small Transformer on modular addition. Save checkpoints. Detect Fourier/cyclic structure. Measure abstraction emergence, obstruction decay, and causal necessity of the cyclic subspace.

3. **IOI component-level calibration.** Use IOI to validate patching infrastructure and compare against attribution patching. If obstruction does not beat attribution, report IOI as an attribution-positive, obstruction-negative control.

4. **Factual recall and ambiguity.** Move the main LLM obstruction claim to tasks where attribution and margin are less saturated, especially margin-matched factual conflicts and ambiguity.

5. **Feature-space extension.** Repeat the strongest experiments with SAE or transcoder features, testing whether feature-space germs produce more stable compatibility structures than residual-space germs.

21 Conclusion

The exact paper goal is to make local-to-global consistency a falsifiable theory of Transformer decision formation.

The exact central contribution is the obstruction-response functional: a corrected obstruction energy coupled to a causal intervention rule.

The center of the original paper is the loop

local evidence → obstruction → predicted intervention → Δobstruction → Δmargin → repair/flip/stabilization.

Supporting material should include sheaf formalism, energy-distance equivalence, Fisher/KL intervention cost, decision-boundary analysis, and carefully scoped failure-mode experiments.

Cut or demote anything that does not improve this loop: decorative field theory, broad thermodynamics, speculative topology, unrestricted tropical geometry, and premature claims about agents or theorem proving.

The sentence to keep in mind while writing the paper is:

> **A theory of AI decisions becomes scientific only when its internal invariant predicts a causal intervention.**

The professor-feedback version adds one prior sentence:

> **A steering theory becomes scientific only after it defines, detects, and causally validates the internal representation being steered.**

The final revised spine is therefore:

representation detection → monitoring → obstruction → response → causal repair.

References

[1] Nanda, N. and Bloom, J. TransformerLens: A Library for Mechanistic Interpretability of Generative Language Models. Software documentation and open-source library.

[2] Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and Editing Factual Associations in GPT. *Advances in Neural Information Processing Systems*, 2022.

[3] Wang, K. R. and colleagues. Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small. 2022.

[4] Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards Automated Circuit Discovery for Mechanistic Interpretability. 2023.

[5] Syed, A., Rager, C., and Conmy, A. Attribution Patching Outperforms Automated Circuit Discovery. 2023.

[6] Geiger, A., Wu, Z., Potts, C., Icard, T., and Goodman, N. Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations. 2021.

[7] Bricken, T. et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. 2023.

[8] Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting Hallucinations in Large Language Models Using Semantic Entropy. *Nature*, 2024.

[9] Yao, S. et al. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. 2023.

[10] Zhang, L., Naitzat, G., and Lim, L.-H. Tropical Geometry of Deep Neural Networks. *International Conference on Machine Learning*, 2018.

[11] Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task. *International Conference on Learning Representations*, 2023.

[12] Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress Measures for Grokking via Mechanistic Interpretability. *International Conference on Learning Representations*, 2023.

[13] Olah, C. et al. Zoom In: An Introduction to Circuits. *Distill*, 2020.

[14] Elhage, N. et al. Toy Models of Superposition. *Transformer Circuits Thread*, 2022.

[15] Placeholder. Neural Networks as Local-to-Global Computations. Add final citation.

[16] Placeholder. Effective-Information Consistency Score for Transformer Circuits. Add final citation.

[17] Placeholder. Transformer Field Theory and Response Functions for Patching. Add final citation.

[18] Placeholder. Persistent Topology of Language Model Representations. Add final citation.